Section 2.2 — Polynomial Arithmetic

Math 103: College Algebra

Adding Polynomials

Add. Simplify your answer as much as possible.

1. $(2m-4)+(5m+1)$
2. $(2w^2-5w-1)+(4w^2-5w+1)$
3. $(4r^2+2.7r-3.6)+(-2r^2-1.3r-4.2)$
4. $\left(3a^2+3a-\frac{5}{6}\right)+\left(2a^3-\frac{2}{3}a+\frac{3}{4}\right)$
5. $\left(\frac{2}{7}ax-\frac{3}{5}ay+\frac{1}{3}axy\right)+\left(\frac{1}{7}ay-\frac{4}{5}axy+\frac{2}{3}ax\right)$
6. $\left(\frac{3}{4}x^4+\frac{1}{3}x^3-\frac{1}{2}x+4\right)+\left(\frac{1}{4}x^4+\frac{2}{3}x^2-\frac{3}{2}x+7\right)$
7. $(x^2y^2+8xy^2-12x^2y-4xy+7y^2-9)+(-3x^2y^2+2xy^2+4x^2y-xy+6y^2+4)$
8. $(-5w^4-3w^3z-8w^2z^2+wz^3-14z^4)+(-3w^4+6w^3z+w^2z^2+12wz^3+5z^4)$

Negating Polynomials

Simplify.

9. $-(2m+3)$
10. $-(8y+5)$
11. $-\left(-5y^2+\frac{1}{5}y-8\right)$
12. $-(-2x^4-3x^2-5)$

Negation and Subtraction

Simplify. Rewrite as subtraction.

Example $-(4-x)$

Solution $-4+x=x-4$

13. $-(x-2)$
14. $-(x^2-y^2)$
15. $-(6-2x)$
16. $-\left(4-\frac{1}{2}z\right)$
17. What pattern do you notice?

Subtracting Polynomials

Subtract. Simplify your answer as much as possible.

18. $(12b^2+4b)-(5b^2+b)$
19. $(7y^2-3y+2)-(5y^2-6)$
20. $(y^2+3y+6)-(-4y^2+2y-10)$
21. $(3u^2v-8uv^2-7v^3)-\left(\frac{1}{3}uv^2+12v^3\right)$
22. $(5x^3-7x^2-2x+3)-(2x+3+5x^3-7x^2)$
23. $(-s^4+7s^3-13s-11)-(3s^4-2s^2-s+5)$
24. $(7.4m^3n-6.7m^2n^2+2.6mn^3)-(3.6m^3n-8.3m^2n^2-4.7mn^3)$

Multiplying by a Monomial

Multiply. Simplify your answer as much as possible.

25. $4n(7n+2)$
26. $-0.4b^2(b^2-3c^2)$
27. $\frac{1}{3}xy^2(2xy-3y^3)$
28. $-5hk^3(3k^3-2h^2k-3hk^4+4h)$
29. $\frac{2}{3}b^4(-30a^2+18ab^3-12a^2b^2)$

Multiplying Binomials

Multiply. Simplify your answer as much as possible.

30. $(a+4)(a+9)$
31. $(2x+4)(4x+3)$
32. $(7y+3x)(2y+4x)$
33. $(a+b)(a-b)$
34. $(xy^2-x^2y)(2xy^2-4xyz)$
35. $(2x+3)^2$

Multiplying Polynomials

Multiply. Simplify your answer as much as possible.

36. $(2x+3)(4x^2+3x-5)$
37. $(u^2-u-1)(u^2+2u+1)$
38. $(x+4)(x^2-4x+16)$
39. $(a^3-a^2+6a+9)^2$
40. $(x+3)^3$
41. $(x+1)(x-2)(x+3)$
42. $(2x-1)^2(x-3)$
43. $(4-x)(2x^2+3x-5)$
44. $(5z^2-2)(2z^2-3z+7)$